Analytical Cume (April 13, 2000)
Theme: Electrochemistry and Fuel Cells

1. (10 points) What is a fuel cell? How does it differ from a combustion engine and from a battery?

2. (15 points) Write the likely chemical half reactions involved in a fuel cell that would be able to power a motor vehicle.

3. (20 points) Given your answer in part ~~(b)~~ 2, above and the table of reduction potentials below, explain why the chemistry that you provided in ~~(b)~~ 2 will produce energy to power the vehicle. Calculate the electromotive force (emf) that would be expected, assuming activities are equal to 0.5.

4. (10 points) Calculate the free energy for this reaction.

5. (5 points) Which is the anode in this reaction? Which is the negative electrode?

6. (10 points) Draw a cell schematic of this electrode cell.

7. (10 points) What are some limitations/hindrances to developing fuel cell technology?

8. (20 points) Describe one electroanalytical technique that you can use to evaluate a a half-reaction? How is the E° obtained from the data? How can the kinetics of the electron transfer be studied? Draw a typical electrochemical response for that half-reaction.

Constant:
F = 96485 Coul/mol

Table C.I
Selected Standard Electrode Potentials in Aqueous Solutions at 25° in V vs. NHE[a]

Reaction	Potential, V
$Ag^+ + e = Ag$	0.7996
$AgBr + e = Ag + Br^-$	0.0713
$AgCl + e = Ag + Cl^-$	0.2223
$AgI + e = Ag + I^-$	−0.1519
$Ag_2O + H_2O + 2e = 2Ag + 2OH^-$	0.342
$Al^{3+} + 3e = Al$(0.1 *M* NaOH)	−1.706
$Au^+ + e = Au$	1.68
$Au^{3+} + 2e = Au^+$	1.29
p-benzoquinone + $2H^+ + 2e$ = hydroquinone	0.6992
$Br_2(aq) + 2e = 2Br^-$	1.087
$Ca^{2+} + 2e = Ca$	−2.76
$Cd^{2+} + 2e = Cd$	−0.4026
$Cd^{2+} + 2e = Cd(Hg)$	−0.3521
$Ce^{4+} + e = Ce^{3+}$(1 *M* H_2SO_4)	1.44
$Cl_2(g) + 2e = 2Cl^-$	1.3583
$HClO + H^+ + e = \frac{1}{2}Cl_2 + H_2O$	1.63

(continued)

[a] More extensive tables can be found in the following sources: (1) W. M. Latimer, "The Oxidation States of the Elements and Their Potentials in Aqueous Solutions," Prentice-Hall, Englewood Cliffs, N.J., 1952 [Note that the signs used by Latimer are opposite to those used here. We follow the convention recommended by the International Union of Pure and Applied Chemistry (IUPAC)]. (2) A. J. Bard and H. Lund, Eds., "The Encyclopedia of the Electrochemistry of the Elements," Marcel Dekker, New York, 1973–. (3) G. Milazzo and S. Caroli, "Tables of Standard Electrode Potentials," Wiley-Interscience, New York, 1977.

An extensive compilation of standard potentials and thermodynamic data under the auspices of the Electrochemistry and Electroanalytical Chemistry Commissions of IUPAC is being prepared (A. J. Bard, J. Jordan, and R. Parsons, Eds.).

Table C.I—(*continued*)

Reaction	Potential, V
$Co^{2+} + 2e = Co$	−0.28
$Co^{3+} + e = Co^{2+}$(3 *M* HNO_3)	1.842
$Cr^{3+} + 2e = Cr$	−0.557
$Cr^{3+} + e = Cr^{2+}$	−0.41
$Cr_2O_7^{2-} + 14H^+ + 6e = 2Cr^{3+} + 7H_2O$	1.33
$Cu^+ + e = Cu$	0.522
$Cu^{2+} + 2CN^- + e = Cu(CN)_2^-$	1.12
$Cu^{2+} + e = Cu^+$	0.158
$Cu^{2+} + 2e = Cu$	0.3402
$Cu^{2+} + 2e = Cu(Hg)$	0.345
$Eu^{3+} + e = Eu^{2+}$	−0.43
$\frac{1}{2}F_2 + H^+ + e = HF$	3.03
$Fe^{2+} + 2e = Fe$	−0.409
$Fe^{3+} + e = Fe^{2+}$(1 *M* HCl)	0.770
$Fe(CN)_6^{3-} + e = Fe(CN)_6^{4-}$(1 *M* H_2SO_4)	0.69
$2H^+ + 2e = H_2$	0.0000
$2H_2O + 2e = H_2 + 2OH^-$	−0.8277
$H_2O_2 + 2H^+ + 2e = 2H_2O$	1.776
$2Hg^{2+} + 2e = Hg_2^{2+}$	0.905
$Hg_2^{2+} + 2e = 2Hg$	0.7961
$Hg_2Cl_2 + 2e = 2Hg + 2Cl^-$	0.2682
$Hg_2Cl_2 + 2e = 2Hg + 2Cl^-$(sat'd. KCl)	0.2415
$HgO + H_2O + 2e = Hg + 2OH^-$	0.0984
$Hg_2SO_4 + 2e = 2Hg + SO_4^{2-}$	0.6158
$I_2 + 2e = 2I^-$	0.535
$I_3^- + 2e^- = 3I^-$	0.5338
$K^+ + e = K$	−2.924
$Li^+ + e = Li$	−3.045
$Mg^{2+} + 2e = Mg$	−2.375
$Mn^{2+} + 2e = Mn$	−1.029
$Mn^{3+} + e = Mn^{2+}$	1.51
$MnO_2 + 4H^+ + 2e = Mn^{2+} + 2H_2O$	1.208
$MnO_4^- + 8H^+ + 5e = Mn^{2+} + 4H_2O$	1.491
$Na^+ + e = Na$	−2.7109
$Ni^{2+} + 2e = Ni$	−0.23
$Ni(OH)_2 + 2e = Ni + 2OH^-$	−0.66
$O_2 + 2H^+ + 2e = H_2O_2$	0.682
$O_2 + 4H^+ + 4e = 2H_2O$	1.229
$O_2 + 2H_2O + 4e = 4OH^-$	0.401
$O_3 + 2H^+ + 2e = O_2 + H_2O$	2.07
$Pb^{2+} + 2e = Pb$	−0.1263
$Pb^{2+} + 2e = Pb(Hg)$	−0.1205
$PbO_2 + 4H^+ + 2e = Pb^{2+} + 2H_2O$	1.46
$PbO_2 + SO_4^{2-} + 4H^+ + 2e = PbSO_4 + 2H_2O$	1.685
$PbSO_4 + 2e = Pb + SO_4^{2-}$	−0.356
$Pd^{2+} + 2e = Pd$	0.83
$Pt^{2+} + 2e = Pt$	~1.2
$PtCl_4^{2-} + 2e = Pt + 4Cl^-$	0.73
$PtCl_6^{2-} + 2e = PtCl_4^{2-} + 2Cl^-$	0.74
$S + 2e = S^{2-}$	−0.508
$Sn^{2+} + 2e = Sn$	−0.1364